

## 11.3 Mode Collapse and Evaluation

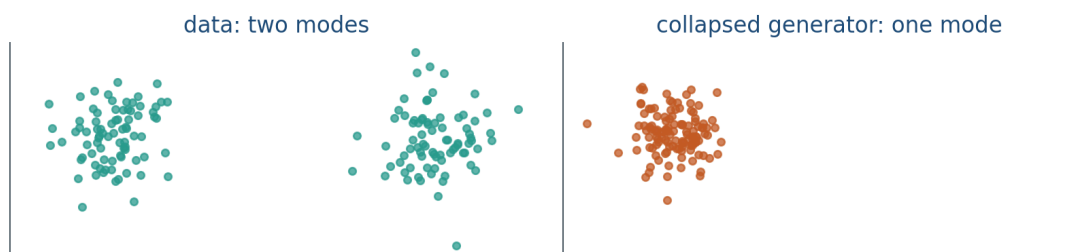
92/8 is collapse. 47/53 is covered.  $G$ 's loss has no missing-mode term.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A GAN has no likelihood you can read as “this model covers the data.” The usual failure is not blur. It is **mode collapse**: the generator finds one (or a few) fakes that fool  $D$  and repeats them.

### 1. One mode is enough to fool $D$

Suppose the reals are two well-separated clusters. If  $G$  parks all of its mass on cluster A,  $D$  learns “cluster A might be fake; cluster B is real.”  $G$  then jumps to B. You cycle. Or  $G$  never leaves A, and every sample looks plausible while half the world is missing.



Collapse is a **coverage** failure. Each fake can look fine. The set of fakes does not.

This is the 1-D spike example from note **11.1** with motion.  $D$  only needs to be wrong on the support of  $G$ . Modes that  $G$  never visits do not appear in  $G$ 's loss at all.

### 2. Quality is not coverage

Eyeballing a grid of images checks **fidelity** (does this one look real?). It does not check whether you missed a mode. Two numbers you want in a report:

| Question                     | Informal name        |
|------------------------------|----------------------|
| Are the fakes realistic?     | precision / fidelity |
| Did we hit the real support? | recall / coverage    |

On images, people cite FID or precision–recall in Inception space. Those are proxies. They need a large sample, and they inherit the classifier's biases. On Lab 11 you can skip Inception: count how many fakes fall near each Gaussian mean.

Latent interpolations are a third check. A short step in  $z$  should change the sample a little. A jump from “shoe” to “face” is a symptom of a broken map, not a feature.

A coverage statistic for two modes: assign each fake to the nearest mean, report the two fractions. Lab 11 calls it collapse if either bin is under 10%. Write both percentages in the report even when they look like 48 / 52: that is the evidence, not a screenshot of two clouds.

---

### 3. What you do about it

There is no single switch. Practical habits: do not let  $D$  become perfect; try more noise dimensions; unrolled or Wasserstein losses in a later paper; **mixture data** in the lab so collapse is visible. For a project, write the coverage check *before* you train. “The pictures look nice” is not a metric.

GANs stay useful for fast sampling. Diffusion (Week 12) is slower and usually covers better, which is why text-to-image moved there. The evaluation lesson does not move: always measure the mode you might have dropped.

---

### 4. Teaching this note

About **30 minutes** at the board, then **~10 minutes** of video. Lab 11 should start in this same class meeting.

- **0–10 min.** Replay the two-spike cartoon. Collapse vs oscillation.
  - **10–20 min.** Fidelity vs coverage table. Write the Lab 11 binning rule on the board.
  - **20–30 min.** Worked count: 100 fakes, 92 / 8. Precision high, recall not.
  - **Then** play CS231N **64:00–74:00** (training is unstable; you do not have  $p(x)$ ; samples can look good anyway). Pause on the “cons” / summary slide and name that gap **coverage**.
- 

### 5. Worked example

Lab 11 means:  $\mu_A = (-2, 0)$ ,  $\mu_B = (2, 0)$ . You sample 100 fakes. Euclidean nearest-mean assignment:

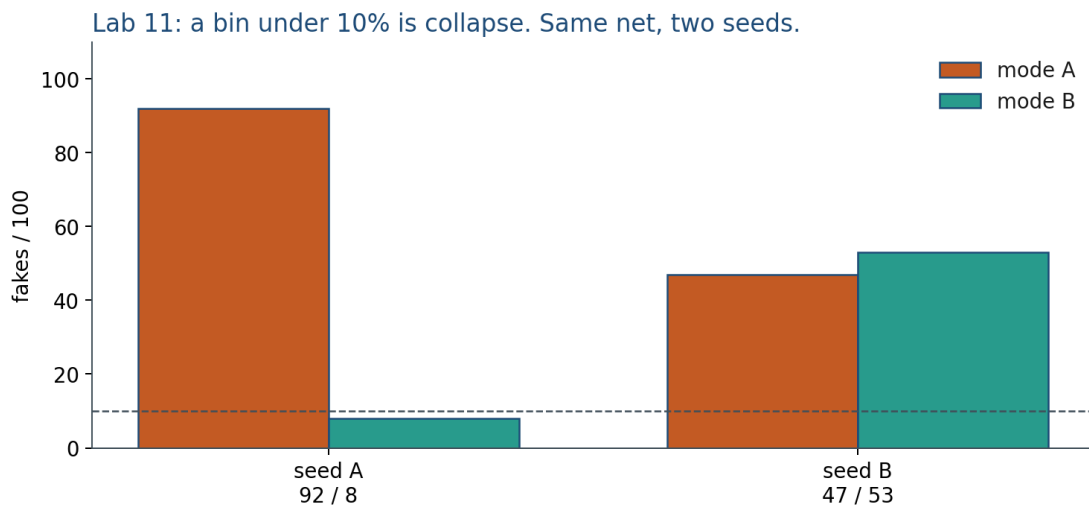
- 92 fakes closer to  $\mu_A$
- 8 closer to  $\mu_B$

Coverage fractions: 0.92 and 0.08. The 8% bin is under 10%: **collapse** (or heavy imbalance) by the lab rule.

A human looking at the 92 points near A says they look like reals. Fidelity is high on A. Recall of the mixture is not.

Generator loss can still be small:  $D$  only fights A, so  $D$  on A drifts toward 0.5, and  $G$  is paid for staying there. The 8 points on B are not a rescue; they are leftover jitter.

If a second seed gives 47 / 53, write **covered**. Same architecture, different outcome. That is why you report the bin fractions, not only `lossG`.



CS231N's GAN recap: you can interpolate in  $z$  when the map is healthy (StyleGAN made that famous). A jump from "shoe" to "face" is a broken map. You will not train StyleGAN here; you will print two bin fractions.

---

## 6. Where students get stuck

- Equating "sharp samples" with a trained GAN. Sharp and identical is collapse.
  - Using  $G$ 's loss as a coverage metric. It has no term for missing modes.
  - Averaging FID (or a 2-D MSE to the origin) across a collapsed cloud and calling it a pass.
- 

## 7. Video

Watch [Stanford CS231N 2017 lecture 13, Generative Models](#), 64:00–74:00.

Pause on instability and the recap that GANs do not give you  $p(x)$  or inference queries. The lecture's word is "unstable"; your word in this course is **mode collapse / coverage**. Same URL as 11.1 and 11.2.

---

## 8. Practice

1. Why can a generator loss look healthy while half the training clusters have no fakes?
2. High precision, low recall: what did the GAN do?
3. You have two 2-D Gaussians as reals. Write one coverage statistic you would print in Lab 11 that is not a loss.
4. 200 fakes; 186 assigned to mean A, 14 to mean B. Do you write collapse under the lab's 10% rule? What are the two fractions?

5.  $D(G(z)) = 0.48$  on every fake, all fakes sit on mode A, and  $D = 0.99$  on mode B. Why is  $G$  not pulled toward B?